Double Hashing

1. Concept

Definition

Double Hashing resolves collisions using two independent hash functions. The second hash function determines the probe step size.

$$h(k,i) = (h_1(k) + i \cdot h_2(k)) \bmod m$$

Where:

- $h_1(k)$ = primary hash
- $h_2(k)$ = step size
- m = table size

$$h_1(k) = k \bmod m$$

$$h_2(k) = 1 + (k \bmod (m-1))$$

$$h(k,i) = (h_1(k) + i \cdot h_2(k)) \bmod m$$

Where:

- k = key
- i = probe number $(i = 0, 1, 2, \dots)$
- m = table size (preferably prime)

For this example:

$$m = 11$$

$$h_1(k) = k \bmod 11$$

$$h_2(k) = 1 + (k \bmod 10)$$

$$\boxed{h(k,i) = (h_1(k) + ih_2(k)) \bmod 11}$$

2. Why This Design?

- $h_2(k) \neq 0$ (guaranteed by $+1$)
- Step size depends on key
- Eliminates primary clustering
- Eliminates secondary clustering
- Ensures full coverage if m is prime

3. Complete Coverage Proof

Theorem

If m is prime and $\gcd(h_2(k), m) = 1$, then the probe sequence visits all m slots.

Assume:

$$h_1(k) + ih_2(k) \equiv h_1(k) + jh_2(k) \pmod{m}$$

Then:

$$(i - j)h_2(k) \equiv 0 \pmod{m}$$

Since m is prime and $h_2(k)$ is relatively prime to m:

$$i = j$$

Hence all probes are distinct.

$$\boxed{\text{Complete Coverage Proven}}$$

4. Simulation Example

Insert:

$$10, 22, 31, 4, 15, 28, 17, 88, 89$$

$$m = 11$$

5. Step-by-Step Insertion

1) Insert 10

$$h_1 = 10, \quad h_2 = 1$$

Index 10 empty $\rightarrow$ Insert.

Probes = 1

2) Insert 22

$$h_1 = 0, \quad h_2 = 3$$

Index 0 empty $\rightarrow$ Insert.

Probes = 1

3) Insert 31

$$h_1 = 9, \quad h_2 = 2$$

Index 9 empty $\rightarrow$ Insert.

Probes = 1

4) Insert 4

$$h_1 = 4, \quad h_2 = 5$$

Index 4 empty $\rightarrow$ Insert.

Probes = 1

5) Insert 15

$$h_1 = 4, \quad h_2 = 6$$

Index 4 occupied

$$(4 + 6) \bmod 11 = 10 \quad \text{(occupied)}$$

$$(4 + 12) \bmod 11 = 5$$

Insert at 5.

Probes = 3

6) Insert 28

$$h_1 = 6, \quad h_2 = 9$$

Index 6 empty → Insert.

Probes = 1

7) Insert 17

$$h_1 = 6, \quad h_2 = 8$$

Index 6 occupied

$$(6 + 8) \bmod 11 = 3$$

Insert at 3.

Probes = 2

8) Insert 88

$$h_1 = 0, \quad h_2 = 9$$

Index 0 occupied

$$(0 + 9) \bmod 11 = 9 \quad \text{(occupied)}$$

$$(0 + 18) \bmod 11 = 7$$

Insert at 7.

Probes = 3

9) Insert 89

$$h_1 = 1, \quad h_2 = 10$$

Index 1 empty → Insert.

Probes = 1

6. Final Insertion Table

Key	Final Index	Probes
10	10	1
22	0	1
31	9	1
4	4	1
15	5	3
28	6	1
17	3	2
88	7	3
89	1	1

$$\boxed{\text{Total Probes} = 14}$$

$$\boxed{\alpha = \frac{9}{11} \approx 0.82}$$

7. Final Hash Table

0	1	2	3	4	5	6	7	8	9	10
22	89		17	4	15	28	88		31	10

8. Time Complexity

$$\alpha = \frac{n}{m}$$

Expected unsuccessful search:

$$E[U] = \frac{1}{1-\alpha}$$

Expected successful search:

$$E[S] = \frac{1}{\alpha} \ln \frac{1}{1-\alpha}$$

Double hashing gives the best distribution among open addressing methods.

Double Hashing = Best Practical Open Addressing Technique